

NAME: Flavie Lavoie-Cardinal

**CURRENT FULL-TIME PRIMARY APPOINTMENT 2023-06-01 AND TITLE: Associate Professor,
Department of Psych**

LANGUAGE SKILLS

LANGUAGE	READ	WRITE	SPEAK	UNDERSTAND	PEER REVIEW
French	Yes	Yes	Yes	Yes	Yes
English	Yes	Yes	Yes	Yes	Yes
German	Yes	Yes	Yes	Yes	Yes

EDUCATION/TRAINING

INSTITUTION AND LOCATION	DEGREE (if applicable)	Completion Date (MM/YYYY)	FIELD OF STUDY
Universität Siegen, Siegen, Germany	BSc	09/2007	Chemistry
Universität Siegen, Siegen, Germany	MSc	11/2008	Chemistry
Universität Siegen, Siegen, Germany	PhD	04/2011	Chemistry
Max Planck Institute for Biophysical Chemistry, Göttingen, Germany	Postdoc	04/2014	Biophysics
Université Laval, Québec, Canada	Postdoc	09/2017	Biophotonics

A. Personal Statement please outline how your training and experience are relevant to the proposed application.

My training and career reflect a sustained commitment to transdisciplinary research and the development of fluorescence-based optical imaging techniques. After earning a PhD in physical and inorganic chemistry from the University of Siegen, Germany, I completed a first postdoctoral fellowship in biophysics (2011–2014) in the laboratory of Prof. Stefan W. Hell, recipient of the 2014 Nobel Prize in Chemistry for the development of super-resolution fluorescence microscopy. There, I contributed to the development of super-resolution optical microscopy techniques. I subsequently joined the laboratory of Prof. Paul De Koninck at Université Laval and the CERVO Brain Research Centre for a second postdoctoral fellowship (2014–2017), before becoming a principal investigator at CERVO (2017–2019). In November 2019, I was appointed Assistant Professor in the Department of Psychiatry and Neuroscience at Université Laval and was promoted to Associate Professor in June 2023. Since 2014, my research has evolved from the development of optical nanoscopy techniques toward their integration with artificial intelligence (AI) to study biological organization at the nanoscale. My team develops AI-assisted microscopy and computational approaches to characterize synaptic organization, diversity, and plasticity. We also design adaptive microscopy strategies that guide image acquisition, with

applications to understanding how aging and chronic stress affect synaptic organization and neuronal function.

In support of my research program at Université Laval, I was awarded the Tier 2 Canada Research Chair in Intelligent Nanoscopy of Cellular Plasticity in 2019, which was renewed in 2024. I hold several scientific leadership positions: I lead the Health and Life Sciences Research Axis at the Institute for Intelligence and Data (IID); co-lead the Perception and Control Axis of the Québec neuroscience and AI research cluster UNIQUE and the Neuroscience and AI Consortium at IVADO; and serve as Scientific Director of both the PREDIS AI for Health Platform and the Canadian Optical Nanoscopy Platform (CONP). Since 2019, I have secured more than \$14 million as principal investigator, including a major Canada Foundation for Innovation grant in 2024 and a Brain Canada Platform Support Grant in 2026 to develop the CONP, as well as more than \$2 million in institutional funding to establish the PREDIS Platform.

A. Professional Experience, Leadership Activities, and Honors

Academic and Scientific Appointments

2023-present **Associate Professor**, Department of Psychiatry and Neuroscience, Université Laval
2019-present **Canada Research Chair Tier 2**, Natural Science and Engineering Research Council
2017-present **Principal Investigator**, CERVO Brain Research Center
2025-present **Associate Academic Member**, Mila
2025-present **Deputy Director**, Cellular and Molecular Neuroscience Research Axis, CERVO Brain Research Center
2025-present **Director**, Centre for Bioimaging at the Nanoscale Abberior-CERVO
2024-present **Scientific Director**, Canadian Optical Nanoscopy Platform
2022-present **Scientific Director**, AI-Health data analysis platform PREDIS, CIUSSS-CN
2020-present **Research Axis Leader**, Health and Life Science Research Axis, Institute for Intelligence and Data
2019-2023 **Assistant Professor**, Department of Psychiatry and Neuroscience, Université Laval

Leadership activities

2026-present **Member**, Scientific Committee, RevoScience
2025-present **Steering Committee Member**, Open Science Office, CERVO Brain Research Center
2024-present **Co-leader**, Neuroscience and AI Research Cluster, IVADO
2022-present **Member**, Equity, Diversity & Inclusion Committee, UNIQUE
2022-present **Member**, Equity, Diversity & Inclusion Committee, CERVO Brain Research Center
2021-present **Research Axis Leader**, FRQNT Strategic Cluster UNIQUE (Union Neurosciences et Intelligence Artificielle Québec)
2020-present **Member**, Scholarship and Funding Committee, UNIQUE
2020-present **Member**, Educational Board, Neurasmus – A European Master in Neuroscience
2019-present **Member**, Ethics Committee, Centre Hospitalier Universitaire de Québec
2022-present **Chair**, Scientific and Organization Committees, Frontiers in Neurophotonics International Summer School

B. Most Important Contributions to Science

1. Machine learning based approach for the online optimization of super-resolution microscopy. We demonstrated the potential of machine learning (ML) for online optimization of super-resolution microscopy. Our approach aims at increasing accessibility of super-resolution microscopy to non-expert users by assisting them in an efficient tuning of imaging parameters, while simultaneously capturing good-quality images. We developed a fully automated optimization platform that combined multi-objective bandits optimization with deep learning (DL) assessment of super-resolution image quality and user preference [13]. This resulted in ML-assisted super-resolution imaging routines that can find optimal acquisition parameters without the need for a user in the loop. This approach can be used to assist non-expert in the optimization of difficult optical

nanoscopy tasks. We demonstrated using a custom-build simulation platform that Reinforcement Learning agents can learn optimal super-resolution imaging trajectory [7].

2. Machine learning strategies for quantitative analysis of microscopy images. I participated to the development and application of several strategies for the quantitative analysis of complex subcellular structures in microscopy images [1,2,5,8-12]. We proposed a latent learning approach, MICRA-Net, that can be trained on a simple main classification task using image-level binary annotations to perform semantic segmentation of biological structures in microscopy images [10]. This approach alleviates the burden of biomedical dataset annotations and increases the accessibility of deep learning assisted microscopy image analysis, especially when expert knowledge is required to perform the annotation task. Using weakly-supervised DL methods, we showed that the F-actin periodical lattice, a structure only visible with super-resolution microscopy, is reorganized to longitudinal fibers in dendrites but not in axons after applying activity promoting stimuli [12]. Our strategy allowed for the first time to reveal an activity-dependent remodeling of the F-actin-based lattice in dendrites. We also developed a multidimensional unsupervised ML framework to characterize activity-dependent modulations of the synaptic organization and morphology [11]. We observed that nanoscale synaptic remodeling is heterogeneous across the synaptic population and often supported by a small number of synaptic subtypes.

3. Deep learning-based approaches for biological discovery. We developed a task-assisted generative adversarial network (TA-GAN) that can use a complementary, biologically relevant task, to generate SR images of neuronal nanostructures from confocal images. When integrated in the live-cell super-resolution image acquisition scheme, the TA-GAN can be used to guide the acquisition process. It helps reducing photobleaching and phototoxicity by imaging only the most informative region with super-resolution and enabling quantitative live-cell & long-term imaging of F-actin nanostructures [11]. Extending this paradigm of biologically informed and task-driven learning, we subsequently developed STED-FM, the first foundation model for STED microscopy, designed to learn transferable representations of nanoscale biological organization and support downstream tasks such as algorithmic super-resolution, denoising, segmentation, and data-driven discovery [1]. By exploring the latent representations learned by STED-FM, we enabled data-driven discovery of subtle and previously overlooked phenotypic changes, including activity-dependent cytoskeletal remodeling and shifts in synaptic nanostructure organization.

4. Super-resolution microscopy technology development. I was involved in the development of novel non-invasive microscopy techniques applied to the investigation of cellular and molecular interactions, which still had a strong impact in the last 6 years in the field of optical nanoscopy, especially regarding its application to live-cell imaging. Altogether the publications in which I was implicated [*Nature* 2011, *ChemPhysChem* 2014, *Science* 2016] were cited more than 750 times, including more than 350 citations in the last 6 years. I contributed to the development RESOLFT (Reversible Saturable Optical Linear Fluorescence Transitions) in living cells, which relied on the use of reversibly switchable fluorescent proteins. We subsequently showed that RESOLFT nanoscopy could be extended to red fluorescent proteins allowing to further decrease phototoxicity by reducing exposition to UV-light. We also demonstrated for the first time two-color RESOLFT nanoscopy using both point scanning and wide-field parallelized illumination methods. The demonstration and applications of RESOLFT nanoscopy in living cells was a major development in the field of super-resolution microscopy. I have also contributed methodological development in fluorescence lifetime imaging (FLIM) for STED microscopy, including applications to FRET and resolution enhancement with SPLIT-STED [4].

5. Application of super-resolution microscopy techniques for nanoscale imaging in the brain. I participated to several transdisciplinary collaboration projects for the application of quantitative super-resolution microscopy to cellular and molecular neuroscience. Our contributions allowed for example to study the how the nanoscale organisation of the cannabinoid receptor CB1 is associated with stress resilience [6], and how autophagy ablations induce cytoskeletal modifications during dopaminergic axon development [3]. In the context of those collaborations, we developed both experimental and analytical approaches that are adapted to the challenges of brain imaging, including imaging in thick brain tissues and quantification of heterogeneous nanostructures.

Selected relevant publications

- [1] Bilodeau, A., Beaupré, F., Chabbert, J., Thériault, K., Bellavance, J.M., Lessard, K., Deschênes, A., Bernatchez, R., De Koninck, P., Gagné, C. and Lavoie-Cardinal, F., (2026). A Self-Supervised Foundation Model for Robust and Generalizable Representation Learning in STED Microscopy *bioRxiv, In Press at Nature Methods*
- [2] Fan, B., Bilodeau, A., Beaupré, F., Wiesner, T., Gagné, C., Lavoie-Cardinal, F. and Hložek, R., (2026). Calcium transient detection and segmentation with the astronomically motivated algorithm for background estimation and transient segmentation (Astro-BEATS) *bioRxiv, Review request at Neurophotonics*
- [3] Profes, M.S., Gora, C., Lavoie-Cardinal, F., Saghatelian, A., Lévesque, M. (2026). Autophagy is required for dopaminergic axon development and confers their responsiveness to guidance cues *Journal of Neuroscience*.
- [4] Deschênes, A., Ollier, A., Lafontaine, M., Michaud-Gagnon, A., Steaven Santiago, J.G., Bilodeau, A., Gagné, C., De Koninck, P. and Lavoie-Cardinal, F. (2025). Enhancing STED microscopy via Fluorescence Lifetime Unmixing and Filtering in Two-Species SPLIT-STED. *Methods in Microscopy*
- [5] Beaupré, F., Bilodeau, A., Wiesner, T., Leclerc, G., Lemieux, M., Nadeau, G., Castonguay, K., Fan B., Labrecque, S., Hložek, R., De Koninck, P., Gagné, C., Lavoie-Cardinal, F. (2025) Quantitative Analysis of Miniature Synaptic Calcium Transients Using Positive Unlabeled Deep Learning. *Digital Discovery*
- [6] Dudek, K., Paton, S.E.J., Collignon, A., Lavoie, O., Bouchard, J., Neutzling Kaufmann, F., Dion-Albert, L., Clavet-Fournier, V., Lebel, M., Manca, C., Flamand, N., Guzman, M., Campbell, M., Turecki, G., Mechawar, N., Lavoie-Cardinal, F., Silvestri, C., Di Marzo, V., Menard, C. (2025) Astrocytic cannabinoid receptor 1 promotes resilience by dampening stress-induced blood-brain barrier alterations. *Nature Neuroscience*
- [7] Bilodeau, A., Michaud-Gagnon, A., Chabbert, J., Turcotte, B., Heine, J., Durand, A., Lavoie-Cardinal, F. (2024) Development of AI-assisted microscopy frameworks through realistic simulation in pySTED *Nature Machine Intelligence*.
- [8] Bouchard, C., Bernatchez, R., Lavoie-Cardinal, F. (2023) Addressing annotation and data scarcity when designing machine learning strategies for neurophotonics. *Neurophotonics*.
- [9] Bouchard, C., Wiesner, T., Deschênes, A., Bilodeau, A., Turcotte, B., Gagné, C., Lavoie-Cardinal, F. (2023) Task-Assisted GAN for Resolution Enhancement and Modality Translation in Fluorescence Microscopy. *Nature Machine Intelligence*
- [10] Bilodeau, A., Delmas, C., Parent, M., De Koninck, P., Durand, A. Lavoie-Cardinal, F. (2022) *Nature Machine Intelligence*
- [11] Wiesner, T., Bilodeau, A., Bernatchez, R., Deschênes, A., Raulier, B., De Koninck, P. and Lavoie-Cardinal, F. (2020) Activity-dependent remodeling of synaptic protein organization revealed by high throughput analysis of STED nanoscopy images. *Frontiers in Neural Circuits*
- [12] Lavoie-Cardinal, F., Bilodeau, A., Lemieux, M., Gardner, M. A., Wiesner, T., Laramée, G., Gagné C., De Koninck, P. (2020) Neuronal activity remodels the F-actin based submembrane lattice in dendrites but not axons of hippocampal neurons. *Scientific Reports*
- [13] Durand, A., Wiesner, T., Gardner, M.A., Robitaille, L.É., Bilodeau, A., Gagné, C., De Koninck, P., Lavoie-Cardinal, F. (2018). A machine learning approach for automated optimization of super-resolution optical microscopy. *Nature Communications*

C. Prior Funding History

Total funding as a principal investigator: **\$14,650,500**

Selected funding (6 ongoing grants out of 38 grants secured as a PI or co-applicant since 2019)

2026-2031 **Principal Investigator**, Pushing the boundaries of super-resolution optical microscopy for molecular imaging in live tissue, NSERC, Discovery Grant, \$245,000

2026-2028 **Nominated Principal Investigator**, Detecting the Undetectable: Shared Signal Processing for Super-Resolution Microscopy and Gravitational Waves, NFRF, Exploration Grant, \$250,000

2026-2029 **Principal Investigator**, Advanced Optical Nanoscopy Platform for Neuroscience, Brain Canada, Platform Support Grant Program, \$1,023,000

2026-2028 **Principal Investigator**, Plateforme de REcherche Données, Intelligence et Santé (PREDIS), Ministère de l'Économie, de l'Innovation et de l'Énergie, PSov2a, \$400,000

2025-2028 **Principal Investigator**, Microscopie multimodale de la structure et la fonction synaptique guidée par l'IA et l'astrophysique, FRQNT/NSERC, NOVA, \$225,000

2023-2028 **Principal Investigator**, Pushing the frontiers of observing nanoscale biology in its native environment, CFI, Innovation, \$7,500,000

D. Activities

Knowledge translation

I have contributed structuring Canada's optical nanoscopy ecosystem through, the establishment of an Abberior-CERVO Excellence Center in 2025, and the creation of the Canadian Optical Nanoscopy Platform for Neuroscience in 2026. I also participated and contributed to the organization of several national teaching initiatives in Optical Nanoscopy, in Québec, Montréal and Calgary. These initiatives broaden access to advanced nanoscopy, AI-assisted microscopy, specialized personnel, and custom experimental protocols. Through PREDIS, I support the integration of AI into health and life-science research across six Québec City research centres; since 2022, the platform has supported more than 75 interdisciplinary projects.

I lead the International Neurophotonics Summer School since 2022 and contribute to Erasmus Mundus Neurasmus program as a member of the Educational Board and organizer of a neurocomputation workshop. I have delivered nine public lectures and contributed to seven publications for general audiences.

Supervisory Activities (Total HQPs 89)

Total HQP scholarships: **\$722,400**

- Research Professionals (12, 10 ongoing)
- Postdoctoral Fellows (2)
- PhD students (12, 8 ongoing)
- MSc students (22, 3 ongoing)
- Undergraduate students (41, 5 ongoing)
- PhD and MSc supervisory or thesis committee (21, 5 ongoing)

Invited talks (out of 69 since 2019)

2026 Invited Speaker, *Foundation models for phenotype discovery in bioimaging*, Institut Pasteur, France

2025 Invited Speaker, *Decoding synaptic diversity with AI-assisted nanoscopy*, MiFoBio, France

2024 Invited Speaker, *Navigating Neuronal Landscapes with Smart Nanoscopy*, EPFL Latsis Symposium on Smart Microscopy, Switzerland

2024 Invited Speaker, *Navigating Neuronal Landscapes with Smart Nanoscopy*, Nature Conference: Transformative Technologies for Neuroengineering, China

2024 Invited Speaker, *Machine Learning Assisted Optical Nanoscopy*. International Conference on Nanoscopy, Germany

2023 Invited Speaker, *Machine Learning-Assisted Super-Resolution Microscopy of Molecular Interactions in the Brain*, Beyond Diffraction: Trends in Nanoscopy Workshop, Portugal